High-Resolution Remote Sensing Bitemporal Image Change Detection Based on Feature Interaction and Multitask Learning

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***Abstract*— With the development of remote sensing technology, high-resolution (HR) remote sensing optical images have gradually become the main source of change detection data. Albeit, the change detection for HR remote sensing images still faces challenges: 1) in complex scenes, a region contains a large amount of semantic information, which makes it difficult to accurately locate the boundaries between different semantics in the feature maps and 2) due to the inability to maintain consistent conditions such as light, weather, and other factors when acquiring bitemporal images, confounding factors such as the style of bitemporal data that are not related to change detection can cause detection difficulties. Therefore, a change detection method based on feature interaction and multitask learning (FMCD) is proposed in this article. To improve the ability to detect changes in complex scenes, FMCD models the context information of features through a multilevel feature interaction module, so as to obtain representative features, and to improve the sensitivity of the model to changes, the interaction between two temporal features is realized through the mix attention block (MAB). In addition, to eliminate the influence of weather and other factors, FMCD adopts a multitask learning strategy, takes domain adaptation as an auxiliary task, and maps the features of bitemporal images to the same space through the feature relationship adaptation module (FRAM) and feature distribution adaptation module (FDAM). Experiments on three datasets show that the proposed method is superior to other state-of-the-art methods.**

***Index Terms*— Change detection, domain adaptation, feature interaction, high-resolution (HR) remote sensing image, multitask learning.**

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I. Introduction

CHANGE detection aims at detecting areas of change between different temporal image pairs and has become a hot topic in remote sensing image processing [1]. This topic is widely applied in many fields, such as environmental monitoring [2], natural disaster assessment [3], urban planning [4], and resource management [5], since it can provide technical reference for decision-making. Automatic acquisition of changes can effectively reduce the consumption of labor and material resources. Therefore, change detection technology based on machine learning or deep learning has received more attention.

In the early research on change detection, researchers use machine learning methods to detect change areas. And these methods processed the independent pixels or neighborhood pixels in the image in a statistical way, such as change vector analysis (CVA) [6] and Markov random field (MRF) models [7]. Albeit, these methods perform poorly in complex scenarios due to poor robustness and insufficient semantic structure.

In recent years, due to its powerful feature extraction ability, deep learning technology has gradually become a hot topic. As the main tool of deep learning, deep neural networks (DNNs), especially convolutional neural network (CNN), can effectively extract features from data. Therefore, more and more scholars have applied it to change detection. The proposal of the Siamese network has brought the field of change detection to a new level by sharing the weight of the network to compare similar parts of bitemporal images [8]. At present, the idea of Siamese networks is widely used in change detection tasks, such as [9], [10], [11], [12], [13], and [14]. Based on the Siamese network, the introduction of the attention mechanism has brought change detection into a new stage, which can increase the receptive field of the model to some extent and improve the ability of the network to extract features in complex scenes. The attention mechanism usually refers to reweighting the resulting feature maps in spatial and channel dimensions such as: [15], [16], [17], [18], and [19].

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Complex scenes make it difficult to accurately locate the boundaries between different semantics in the feature maps, so extracting representative features is of great significance for change detection. P2V-CD [20] locates the change area through the time correlation of two temporal images. Since the transformer can extract long-range semantic information, it is widely used in image classification [21], image segmentation [22], object detection [23], and so on. Li et al. [24] and Chen et al. [25] combine transformer and change detection to analyze the change information of the images from a global perspective, which achieves good results.

However, the above methods operate on the high-level features of images at a very deep stage. For change detection, the high-level features of the image can provide richer semantic information and the low-level features can reflect the differences in the time domain [12]. Therefore, modeling the context information of features and aggregating the features between two temporal images can improve the ability of the model to detect changes. In other words, feature interaction is the key to detecting changes.

What's more, because of the particularity that data of change detection is pairwise, the problem of inconsistent conditions when acquiring bitemporal images is inevitable, which directly leads to an inconsistent style of bitemporal images. So, change detection focus on the difference of related factors such as ground object content, and the influence of confounding factors such as style should be reduced as much as possible. The difference in style between bitemporal images is essentially the divergency in the domain of the image data, that is domain shift. To solve the problem of domain shift, the domain adaptation technique is introduced to narrow the data distribution distance between the source domain and the target domain. With the introduction of adversarial learning, generative adversarial network (GAN) can effectively learn data distribution implicitly and is widely used in the field of domain adaptation such as: [26], [27], and [28]. Based on this principle, Liu et al. [29] learns the distribution of data implicitly between source and target domains through adversarial learning to realize the style transfer between two temporal images and proposed a domain adaptive change detection framework, which has achieved remarkable results. Albeit, a large number of parameters for generating adversarial networks makes the model difficult to train.

In summary, there are two main challenges in change detection for high-resolution (HR) remote sensing images: 1) changes in complex scenarios are difficult to detect and 2) it is a predicament to guarantee the consistency of conditions when acquiring bitemporal data. As shown in Fig. 1, due to the difference in sensors, weather, light, and other factors, the style of bitemporal images becomes discrepant, which makes it difficult to detect change areas. Meanwhile, some irrelevant changes are introduced into the bitemporal images. The yellow box represents pseudo changes in shadow due to differences in light. Pseudo changes like this increase the complexity of the scene, making change detection difficult.

Therefore, a change detection method based on feature interaction and multitask learning (FMCD) is proposed in this article. FMCD extracts representative features through the interaction of features at different levels of a temporal and the interaction of features between two temporals. And a multitask learning strategy is adopted in FMCD, that is, a domain-adaptive auxiliary task is set up to align features between two temporal images to improve the performance of the main task of change detection [30]. The two tasks share weights in the feature encoder part and then process the features from the feature encoder separately. In the feature encoder part, a multilevel feature interaction module models the context information of one temporal feature. In the change detection task, a mix attention block (MAB) is proposed. MAB can not only make full use of the spatial–spectral joint features of multiple scales but also aggregate features between two temporal images, which improves sensitivity to change areas. In the domain adaptation task, irrelevant changes due to inconsistent conditions are mainly reflected in low-level features, so, to solve the problem, the feature relationship between the two temporal images is first aligned. After the multilevel features are aggregated in the multilevel feature interaction module. The feature distribution of the two temporal images is aligned by an autoencoder in a reconstructed way, aiming at mapping the two temporal features into the same space.



Fig. 1. Illustration of inconsistent conditions when acquiring bitemporal images. The first image is the pre-event image, the second image is the post-event image, and the third image is the change ground-truth map. The bitemporal images are taken in 2012 and 2016, respectively. And the yellow box represents pseudo changes in shadow due to differences in light.

The main contributions are as follows.

1) Aiming at improving the ability of the model to detect changes in complex scenes through feature interaction, a multilevel feature interaction module is used to interact with different levels of features in one temporal, and a new skip connection block MAB is proposed to interact with different temporal features.
2) A multitask learning strategy is adopted in FMCD. The change detection is set as the main task and the domain adaptation between two temporal images is set as an auxiliary task. By setting the auxiliary task, the influence of confounding factors such as style and irrelevant changes on the change detection task is weakened, so as to improve the accuracy and robustness of the model.
3) The domain adaptation between two temporal images is realized by aligning the feature relationship and feature distribution. The feature distribution adaptive module maps the features of the bitemporal images to the same space by aligning the distribution of the reconstructed images, thereby improving the accuracy of the detection. The feature relationship adaptive module aims at reducing the interference of irrelevant changes by aligning the relationship between the two temporal features.

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The rest of this article is organized as follows. Section II introduces the details of FMCD. Experimental results and analysis are presented in Section III, and Section IV gives the conclusion.

II. Related Works

A. Deep Learning-Based Change Detection

In recent years, the use of DNN has made a leap in the development of change detection. Daudt et al. [8] proposes Siamese networks, which can compare two images by sharing network weights. In the Siamese networks, two temporal images pass through the same network, which not only shortens the distance between similar regions but also reduces the number of network parameters. Based on this principle, Zhang et al. [10] and Shi et al. [14] integrate the deep supervision strategy into the decoder part of the Siamese network, which can constrain the ability of the network to detect changes at the label level. Starting from improving the feature extraction ability of the network, Chen et al. [9] and Huang et al. [11] add multiscale convolution units and attention modules to the encoder part, respectively. P2V-CD [20] integrates video understanding tasks into change detection to extract temporal and spatial information. TINYCD directly shares the weight of the existing network to achieve the purpose of the Siamese network [12]. Yang et al. [13] shares the deep network weights of the encoder to achieve semantic-level change detection. These methods all have achieved good results.

With the development of the attention mechanism, more and more networks indirectly expand the receptive field of the network through the attention mechanism, thereby improving the performance of the network. SNUnet [15] realizes the channel attention mechanism by assigning different weights to different channels in the feature map. Chen and Shi [16] reweight the spatial structure in the feature map by pixel-level convolution to achieve spatial attention. Lv et al. [17], Feng et al. [18], and Liu et al. [19] combine spatial attention and channel attention so that the network can improve the detection ability to change regions at different angles. Although the above methods expand the receptive field of the network to a certain extent, it is still difficult to obtain the long-range dependence of features. Transformer can effectively extract global features through its self-attention mechanism, so Li et al. [24] designed the transformer module in the skip-connection part of the U-net to obtain global information at different scales. To solve the problem of excessive computation in transformer, Chen et al. [25] proposed a bitemporal image transformer (BIT), which converted feature maps of bitemporal images into semantic tokens.

B. Domain Adaption

Domain adaptation is a branch of transfer learning. Its essence is to map two data belonging to different domains into the same feature space so that the data of the two domains can be calculated [26]. Generally speaking, there are two main ways to realize domain adaptation: the reconstruction-based method and the adversarial-based method.

The adversarial-based method is mainly realized by GAN. The generator in GAN generates a feature, and then the discriminator distinguishes whether the feature belongs to the source domain or the target domain, and implicitly learns the data distribution of the source domain and the target domain by adversarial training. Tzeng et al. [26] realizes the mutual recognition between different styles of numbers through GAN. Due to the instability of GAN training, Korotin et al. [27] proposed W2GN, which measures the difference between the two domains by Wasserstein-2 distance. Based on the idea of style transfer, Karras et al. [28] proposed StyleGAN, which redesigned the architecture of the generator and realized the style conversion between two domains of images by controlling the image synthesis process. In the field of change detection, Liu et al. [29] implicitly learns the style of the image between two temporal through StyleGAN, and finally, the images after style transformation are sent as the input of the change detection network, which achieves good results.

The reconstruction-based methods generally use an encoder–decoder structure. In [31], the source domain data and the target domain data share the same encoder, where the source domain features are classified by a classifier, and the target domain features are decoded and reconstructed by a decoder. Generally speaking, the reconstruction-based method requires fewer parameters than GAN. And in change detection, the data of change detection are pairwise, so the difference in content between the data in the source domain and data in the target domain has little effect on the adaptive results. Therefore, considering the parameters of the model and the characteristics of the change detection data, FMCD uses the reconstruction-based method to realize the domain adaptation between the two temporals, that is, the data in the source domain and the data in the target domain will be reconstructed in the decoder.

C. Multitask Learning

Multitask learning is a kind of joint learning, which means that multiple tasks learn together and the results influence each other. In deep learning, there are two ways to implement multitask learning, namely hard parameter sharing and soft parameter sharing. The main difference between the two methods is the way of parameter sharing. Hard parameter sharing is to share all the underlying parameters, and the top-level parameters are independent of each task [32]. Soft parameter sharing is that each task shares part of the parameters at the bottom, and the other part is not shared [33]. The top-level parameters are independent. Although there is only one change detection task in change detection, setting appropriate auxiliary tasks can improve the learning performance of the main task [34]. For change detection, the same domain of two temporal data, that is, the same time, illumination, climate, and other factors can effectively improve the detection ability of the changing area. Therefore, in this article, the domain adaptation between two temporal images is set as an auxiliary task of the change detection task.

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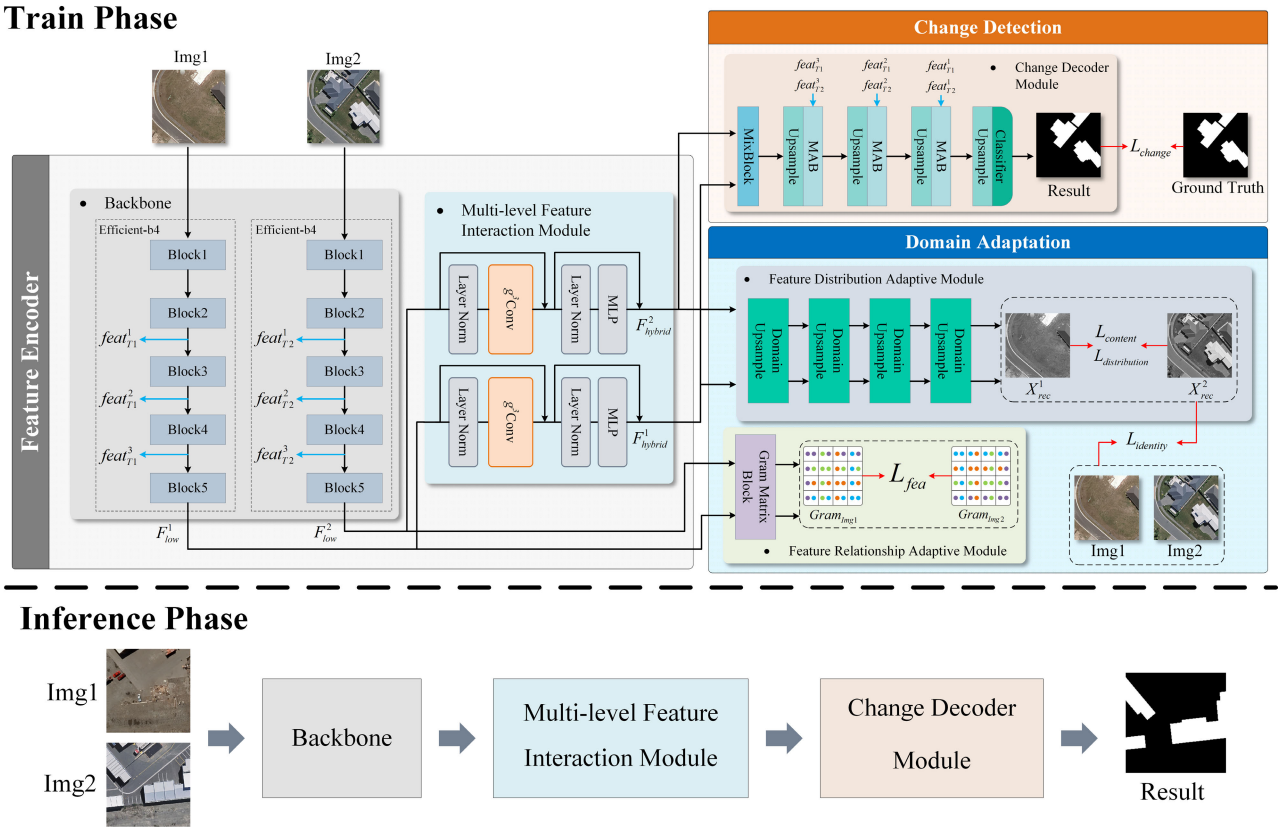


Fig. 2. Framework of FMCD.

III. Proposed Method

In this section, the overall framework of the FMCD is first introduced. Then a presentation of the strategy of multitask learning is given. Next, the feature encoder is explained in detail. Finally, the change detection task and the domain adaptation task are introduced separately.

A. Overview

As shown in Fig. 2, FMCD with a multitask learning strategy consists of three parts: feature encoder, change detection, and domain adaptation. The change detection task and the domain adaptation task share weights in the feature encoder part, and then process the features from the feature encoder separately [35], [36].

The feature encoder part contains two modules: the backbone and the multilevel feature interaction module. In the feature encoder part, assuming that the input images are $\mathcal{X}^1, \mathcal{X}^2 \in \mathbb{R}^{H\times W\times 3}$, where H and W are the height and width of the images, respectively. $\mathcal{X}^1$ and $\mathcal{X}^2$ are first extracted as low-level features $\mathcal{F}^1_{\text{low}}, \mathcal{F}^2_{\text{low}} \in \mathbb{R}^{H_0\times W_0\times C_0}$ via backbone with pretrained weights, where H_0, W_0, and C_0 are the height, width, and channels of the features. Then $\mathcal{F}^1_{\text{low}}$ and $\mathcal{F}^2_{\text{low}}$ are sent to the multilevel feature interaction module where the low-level features can be interactively aggregated with the high-level features to extract more representative hybrid features $\mathcal{F}^1_{\text{hybrid}}, \mathcal{F}^2_{\text{hybrid}} \in \mathbb{R}^{H_0\times W_0\times C_0}$.

In the change detection task, $\mathcal{F}^1_{\text{hybrid}}$ and $\mathcal{F}^2_{\text{hybrid}}$ are merged through the mix block, then gradually upsampled and classified to obtain the final result $\mathcal{X}_{\text{result}} \in \mathbb{R}^{H\times W\times 1}$ in the change detection module. During the upsampling process, multiscale features from the backbone are introduced into the upsampled backbone via MAB as skip connections [37], [38].

In the domain adaptation task, the feature relationship adaptive module and feature distribution adaptive module are adopted. To alleviate the irrelevant interference caused by the discrepancy of weather and illumination, the feature relationship adaptive module aligns the feature relationship between the low-level features $\mathcal{F}^1_{\text{low}}$ and $\mathcal{F}^2_{\text{low}}$ by aligning their gram matrix. Then, the contents of $\mathcal{F}^1_{\text{hybrid}}$ and $\mathcal{F}^2_{\text{hybrid}}$ are reconstructed through the feature distribution adaptive module. And then the distribution of the data tends to be uniform, so as to achieve the purpose of mapping the features of the bitemporal images into the same space.

After the model is trained, in the inference stage of the model, the input images only need to go through the backbone, multilevel feature interaction module, and change detection module to get the final change detection result.

B. Strategy of Multitask Learning

As an auxiliary task of the change detection task, the domain adaptation task optimizes the results of change detection in another direction to improve the performance of the change detection task [34]. In FMCD, the change detection task is essentially to detect the change category of pixels, which is a binary classification task. The domain adaptation task is equivalent to aligning the features between two temporals,

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so it belongs to the regression task. Two tasks have different magnitudes of loss due to different categories, so the FMCD loss is generally shown in the following formula:

$$L_{\text{total}} = w_c L_{\text{change}} + w_d L_{\text{domain}} \tag{1}$$

where L_{change} and L_{domain} are the loss of change detection task and domain adaptation task, respectively, w_c and w_d are the loss weights of the two tasks. The general way to measure the importance of the loss of different tasks is to set a weighting factor, as shown in the following formula:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{domain}} + (1-\alpha)\mathcal{L}_{\text{change}}. \tag{2}$$

However, the weights of loss for the two tasks are generally different for different datasets. Considering that the domain adaptation task and the change detection task use the same dataset, the method we proposed uses the homoscedasticity uncertainty in the deep learning model to adaptively measure the weight of the two task losses [39]. As shown in the following formula:

$$L_{\text{total}} = \frac{1}{2\sigma_1^2} L_{\text{change}}(W) + \frac{1}{2\sigma_2^2} L_{\text{domain}}(W) + \ln \sigma_1 + \ln \sigma_2 \tag{3}$$

where W represents the weight of the network model, σ_1 and σ_2 represent the observation noise and are learnable parameters that measure the weight of losses of the two tasks. The following logarithmic functions in the formula (3) are the regular terms corresponding to each loss function. To ensure that the regular terms of each loss are not negative, Liebel and Körner [40] modifies the regular terms in the loss function, and the final loss function is shown in the following formula:

$$L_{\text{total}} = \frac{1}{2\sigma_1^2} L_{\text{change}}(W) + \frac{1}{2\sigma_2^2} L_{\text{domain}}(W) + \ln\left(1+\sigma_1^2\right) + \ln\left(1+\sigma_2^2\right). \tag{4}$$

C. Feature Encoder

The feature extractor consists of two parts, one is the backbone, and the other is a multilevel feature interaction module. As for the backbone, the initial goal is to obtain a network with the ability to extract shallow features, but due to the small amount of data in the change detection dataset, it is difficult to obtain a network with better shallow feature extraction through training, so the network with pretraining weights is considered. While the pretrained model is not trained on the data used by the change detection task, the processing of low-level features such as color, texture, and so on is similar and generic [41]. The EfficientNet family is efficient and effective because of its model composite scaling method [42]. Therefore, the first five blocks of EfficientNet_b4 with pretrained weights on the ImageNet dataset are selected as the backbone of the FMCD. In the backbone, the bitemporal images $\mathcal{X}^1$ and $\mathcal{X}^2$ pass through an Efficient_b4 network, respectively. In the process of extracting low-level features, the results of the last three blocks are retained as the input of the subsequent MAB.

In the multilevel feature interaction module, different levels of features are aggregated to obtain more representative features. Therefore, two HorNets [43] are used as the multilevel feature interaction module in this article. HorNet has made great progress in the field of image processing due to its ability to realize the interaction between multilevel spaces. The structure of the module is shown in Fig. 2. The HorNet we adopted emulates the structure of the transformer by replacing the multihead self-attention layer with the three-order recursive gated convolution (g^3Conv). The input data are passed through a normalization layer (LayerNorm) before passing through g^3Conv and multilayer perceptron (MLP). And the MLP layer consists of a fully connected layer with an expansion factor of 4. Both g^3Conv and the self-attention mechanism interact with features. The self-attention mechanism interacts with its own features to calculate the attention score, while g^3Conv interacts with three levels of features. The general structure of g^3Conv is shown in Fig. 3. Assuming that the input feature map is $\mathcal{F}_{\text{low}} \in \mathbb{R}^{H_0 \times W_0 \times C_0}$. First, $\mathcal{F}_{\text{low}}$ passes through a convolutional layer, mapping the channel to $2C_0$. Then the obtained feature map is divided into two parts $\mathcal{F}_{\text{part1}} \in \mathbb{R}^{H_0 \times W_0 \times (C_0/2^2)}$ and $\mathcal{F}_{\text{part2}} \in \mathbb{R}^{H_0 \times W_0 \times (2C_0 - (C_0/2^2))}$, according to the number of levels i. The information in $\mathcal{F}_{\text{part2}}$ is then reaggregated by a depth-wise convolution layer and divided into n parts

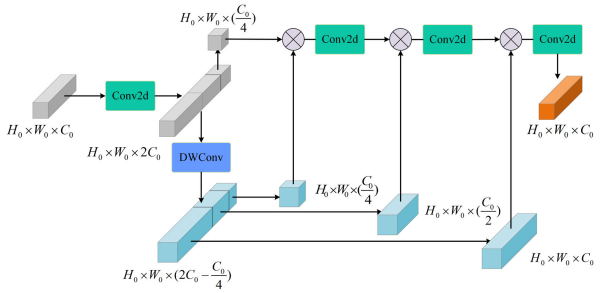


Fig. 3. Illustration of g^3Conv.

$$\mathcal{F}_{\text{part2}}^j \in \mathbb{R}^{H_0 \times W_0 \times \frac{C_0}{2^{3-j}}}, \quad j = 1, 2, 3. \tag{5}$$

Finally, the output $\mathcal{F}_{\text{hybrid}}^j$ of the module is shown in the following formulas:

$$\mathcal{F}_{\text{hybrid}}^{ij} = \begin{cases} \text{Conv2d}\left(\mathcal{F}_{\text{part1}} * \mathcal{F}_{\text{part2}}^j\right), & j = 1 \\ \text{Conv2d}\left(\mathcal{F}_{\text{hybrid}}^{ij-1} * \mathcal{F}_{\text{part2}}^j\right), & j = 2, 3 \end{cases} \tag{6}$$

$$\mathcal{F}_{\text{hybrid}}^i = \mathcal{F}_{\text{hybrid}}^{i3}, \quad i = 1, 2. \tag{7}$$

Conv2d is a pixel-level convolution layer, which doubles the number of input channels to accommodate the next operation. $*$ represents the multiplication of the corresponding elements of the matrix.

D. Change Detection Task

As for change detection tasks, the feature interaction between two temporal images can model the change information from the perspective of the time domain, which is very important [44]. Therefore, a feature mixing method is used in the change detection module, that is, the mix block in Fig. 2.

The detailed process in the mix block is shown in Fig. 4. In the mix block, the two temporal inputs Input1, Input2 $\in \mathbb{R}^{H_0 \times W_0 \times C_0}$ cross each other in the channel dimension to obtain a new feature map Fea $\in \mathbb{R}^{H_0 \times W_0 \times 2C_0}$, so that the information

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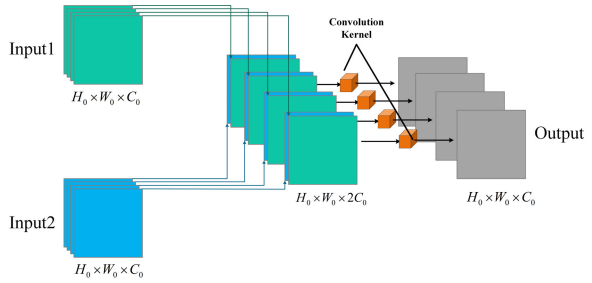


Fig. 4. Illustration of mix block.

can be fused in the temporal dimension. Then, in pairs on the channel dimension, Fea is divided into C_0 groups, and each group aggregates the information through a convolution kernel to generate mix block output Output $\in \mathbb{R}^{H_0 \times W_0 \times C_0}$. The entire operation of the mix block can be implemented by the group parameter in Conv2d in Pytorch [12], as shown in the formula

$$\begin{aligned}\text{Output} = \text{PReLU}(\text{IN2d}(\text{Conv2d}(\text{concat}(\text{Input1},\text{Input2})\\ \text{ch}_{\text{in}} = 2C_0, \text{ch}_{\text{out}} = C_0, \text{groups} = C_0)))\end{aligned} \tag{8}$$

where PReLU is the activation function layer, IN2d is InstanceNorm2d(), and concat is the concatenation operation. It is worth noting that the working principle of mix block is to interact with each channel of the two feature maps. Therefore, InstanceNorm2d() is more suitable for mix block, whose normalized statistics, that is, the mean and variance, are calculated based on the number of channels in the feature map.

In the process of upsampling, an attention module, MAB, for skip connection is proposed. As shown in Fig. 5, the bitemporal multiscale feature maps from the backbone are mixed by a mix block and generate a change attention map by a change attention block. The output of the mix block is then mixed with the output from the previous upsampling phase and finally multiplied by the change attention map to get the input from the new upsampling phase. As shown in the following formulas:

$$\text{CAttn}^i = \text{CAB}\left(\text{feat}_{T1}^i, \text{feat}_{T2}^i\right), \quad i = 1, 2, 3 \tag{9}$$

$$\text{Up}^i = \begin{cases} \text{Mix}\left(\mathcal{F}_{\text{hybrid}}^{T1}, \mathcal{F}_{\text{hybrid}}^{T2}\right), & i = 3 \\ \text{Mix}\left(\text{Upsample}\left(\text{Up}^{i+1}\right), \text{Mix}\left(\text{feat}_{T1}^{i+1}, \text{feat}_{T2}^{i+1}\right)\right) \\ \quad \times \text{CAttn}^{i+1}, & i = 2, 1, 0 \end{cases} \tag{10}$$

where CAB stands for the change attention block, Mix represents the mix block, feat_{T1}^i and feat_{T2}^i stand for bitemporal multiscale feature maps extracted from backbone, respectively. Upsample represents the upsampling operation, which is done by a transposed convolutional layer with a kernel size of 3.

As for change attention block, inputs feat_{T1}^i, feat_{T2}^i are first flattened and normalized for subsequent calculations

$$V_{1,2}^i = L2\text{Norm}\left(\text{flatten}\left(\text{feat}_{T1,T2}^i\right)\right). \tag{11}$$

Since cosine similarity pays more attention to the difference between two vectors in direction, rather than distance or length, and in the change detection task, the main characteristic of the feature vector corresponding to the change region is the discrepancy in direction, so cosine similarity is adopted in this article. The calculation method is shown in the following formula:

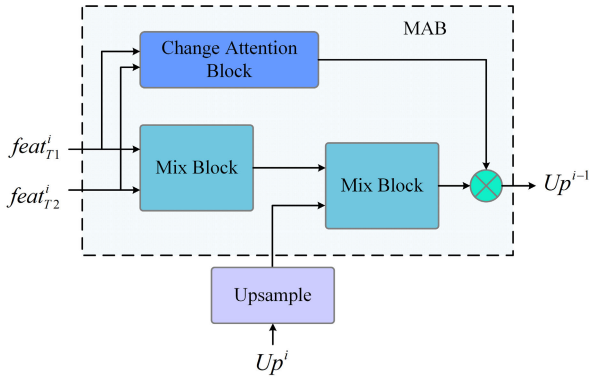


Fig. 5. Illustration of MAB and upsample.

$$\text{Sim}^i = \frac{V_1^i \cdot V_2^i}{\|V_1^i\| \cdot \|V_2^i\|}. \tag{12}$$

The more similar the vectors are, the closer their cosine similarity value is to 1. However, in the task of change detection, the more dissimilar the vector pairs, the more likely they are to change and the closer they are to 1 in the feature map. So the result needs to be processed to obtain the final change attention map, as shown in the following formula:

$$\text{CAttn}^i = 1 - \text{Sigmoid}(\text{Sim}^i). \tag{13}$$

The prediction Result is obtained by classifying the upsampled results by a pixel-wise classifier which consists of three convolutional layers with a kernel size of 1.

Finally, the loss of change detection is calculated by a binary cross entropy loss function, as shown in the following formula:

$$L_{\text{change}} = \text{BCELoss}(\text{Result, Groundtruth}). \tag{14}$$

E. Domain Adaptation Task

The data of change detection are pair-wise and the image content of each pair of two temporal images is similar. Therefore, in the domain adaptation task, each pixel in the image is treated as a sample, and the domain distribution between the two temporal images is aligned by aligning the distribution of the samples in each pair of images. This is different from general domain adaptation, which treats each image as a sample. In the domain adaptation task, two adaptive alignment modules are adopted to align the feature of bitemporal images, which are the feature relationship adaptive module and the feature distribution adaptive module.

The feature relationship adaptive module first obtains an affinity matrix by calculating the similarity between the features in the feature map. And then, the feature relationships of the two temporals are aligned by aligning the corresponding affinity matrices, so as to reduce the impact of irrelevant changes on detection. The Gram matrix [45], [46] is used to measure the characteristics of each dimension and the relationship between dimensions, which is chosen as the core of the feature relationship adaptive module. In this module, the Gram matrix extracts the hidden connections between features

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by calculating the correlation between the shallow features extracted by backbone. For the Gram matrix module input $\mathcal{F}_{\text{low}} \in \mathbb{R}^{H_0 \times W_0 \times C_0}$, first flatten it into $\mathcal{F}^g \in \mathbb{R}^{C_0 \times W_0 H_0}$, then the gram matrix is calculated as shown in the following formula:

$$\text{Gram}_{ij} = \sum_{k}^{W_0 H_0} \mathcal{F}^g_{ik} \mathcal{F}^g_{jk}, \quad i, j = 1, 2, \ldots, C_0 \tag{15}$$

where $\mathcal{F}^g_{ik}$ is the eigenvector of $\mathcal{F}^g$, and $\mathcal{F}^g_{jk}$ is the eigenvector of the transpose matrix of $\mathcal{F}^g$. After obtaining the Gram matrix of the low-level features of the bitemporal image, the loss of the Gram matrix of the bitemporal image is calculated to measure the correlation and compactness of the features between the bitemporal images. The calculation formula is as follows:

$$L_{\text{fea}} = \frac{1}{C_0^2} \sum_{i,j} \left(\text{Gram}^1_{ij} - \text{Gram}^2_{ij}\right)^2. \tag{16}$$

After the relationship alignment of low-level features, the low-level features and high-level features are aggregated by the multilevel feature interaction module to obtain representative features. Mapping these features that aggregate context information to the same space helps improve the accuracy of change detection. In the feature distribution adaptive module, a simple decoder is used to reconstruct bitemporal images on the image content. In FMCD, a simple reconstruction decoder is designed to match the complex and powerful feature encoder. A subjective idea is that when the parameters of the decoder are small, to complete the reconstruction task, the feature encoder with a complex structure will be trained to learn more information. In the domain adaptation task, the main work of the reconstruction decoder is to narrow the distance between the source domain and the target domain and maps the features of the two temporal images to the same feature space. Therefore, designing a simple and weak reconstruction decoder can prompt the feature encoder to generate representative features in the same feature space. And in this process, the bitemporal images share the weight in the module so that the feature of the bitemporal images are mapped to a unified space. As shown in the domain adaptive part of Fig. 2. Each domain upsampling module consists of three convolutional layers and a transposed convolutional layer. The kernel size of the convolutional layer is 3, 3 and 1, and a PReLU() activation function is set at the back of each convolutional layer. Due to [47] shows that the regularization layer will affect the reconstruction effect, so unlike previous experience, the regularization layer is not set. The operation process is shown in the following formula:

$$\mathcal{F}^i_{\text{up}} = \begin{cases} \mathcal{F}_{\text{hybrid}}, & i = 1 \\ \text{Conv}_{\times 3}\left(\text{TransConv}\left(\mathcal{F}^{i-1}_{\text{up}}\right)\right), & i = 2, 3, 4 \end{cases} \tag{17}$$

$$\mathcal{X}_{\text{rec}} = \mathcal{F}^4_{\text{up}}. \tag{18}$$

For the outputs $\mathcal{X}^1_{\text{rec}}, \mathcal{X}^2_{\text{rec}} \in \mathbb{R}^{H \times W \times 3}$ obtained in the feature distribution adaptive module, the reconstruction loss between them and the bitemporal images $\mathcal{X}^1, \mathcal{X}^2$ is first obtained by calculating the content consistency loss

$$\mathcal{L}_{\text{identity}} = \sum_{i=1}^{2} \left\| \text{Norm}\left(\mathcal{X}^i_{\text{rec}}\right) - \text{Norm}(\mathcal{X}^i) \right\|_2 \tag{19}$$

where Norm denotes the mean–variance channel-wise normalization, then the content loss and feature distribution loss between $\mathcal{X}^1_{\text{rec}}$ and $\mathcal{X}^2_{\text{rec}}$ are calculated. For content loss, because the two outputs are from the input of different times, there are inevitably differences in content. Therefore, the change ground-truth map $\text{CM} \in \mathbb{R}^{H \times W \times 1}$ is introduced as a mask to calculate their content loss. The formula is as follows:

$$\mathcal{L}_{\text{content}} = \left\| \text{Norm}\left(\mathcal{X}^1_{\text{rec}}(1 - \text{CM})\right) - \text{Norm}\left(\mathcal{X}^2_{\text{rec}}(1 - \text{CM})\right) \right\|_2. \tag{20}$$

At the same time, to map the features of the bitemporal image in the same space, the data distributions of $\mathcal{X}^1_{\text{rec}}$ and $\mathcal{X}^2_{\text{rec}}$ need to be aligned as follows:

$$\mathcal{L}_{\text{distribution}} = \left\| \mu\left(\mathcal{X}^1_{\text{rec}}\right) - \mu\left(\mathcal{X}^2_{\text{rec}}\right) \right\|_2 + \left\| \sigma\left(\mathcal{X}^1_{\text{rec}}\right) - \sigma\left(\mathcal{X}^2_{\text{rec}}\right) \right\|_2 \tag{21}$$

where μ represents the mean value of the value on the feature map, and σ represents the variance. In this task, $\mathcal{L}_{\text{fea}}$, $\mathcal{L}_{\text{identity}}$, $\mathcal{L}_{\text{content}}$, and $\mathcal{L}_{\text{distribution}}$ are all calculated by mean square error (mse). Because the calculation method of these losses is similar, the final domain adaptive loss is obtained by directly adding them. The formula is as follows:

$$\mathcal{L}_{\text{domain}} = \mathcal{L}_{\text{fea}} + \mathcal{L}_{\text{identity}} + \mathcal{L}_{\text{content}} + \mathcal{L}_{\text{distribution}}. \tag{22}$$

IV. Experiments

In this section, the HR optical remote sensing image datasets used in the experiment for change detection are introduced first. Then, the implementation details of experiments are given. Finally, experimental results and ablation analysis of each module in FMCD are given.

A. Datasets and Evaluation Metrics

In this article, we tested the proposed method on three datasets, namely WuHan University (WHU-CD) [48], SYSU-CD [14], and the deeply supervised image fusion network dataset (DSIFN-CD) [10].

WHU-CD is a dataset for building changes, with a spatial resolution of 0.2 m. The total size of the image is 32 507 × 15 354. In this article, the original images are cut into small patches of size 256 × 256 with no overlap, and then, they are randomly divided into the training set, validation set, and test set, with the number of 5950/742/742.

SYSU-CD contains 20 000 pairs of images taken in Hong Kong in 2007 and 2014. The main changes in the dataset are reflected in terrain changes such as buildings, vegetation, and roads. The resolution of this dataset is 0.5 m. The author of this dataset crops the data into image patches of size 256 × 256 and splits them into the training set, validation set, and test set in a scale of 6:2:2.

DSIFN-CD is derived from Google Earth and covers six cities in China, namely Beijing, Chengdu, Shenzhen,

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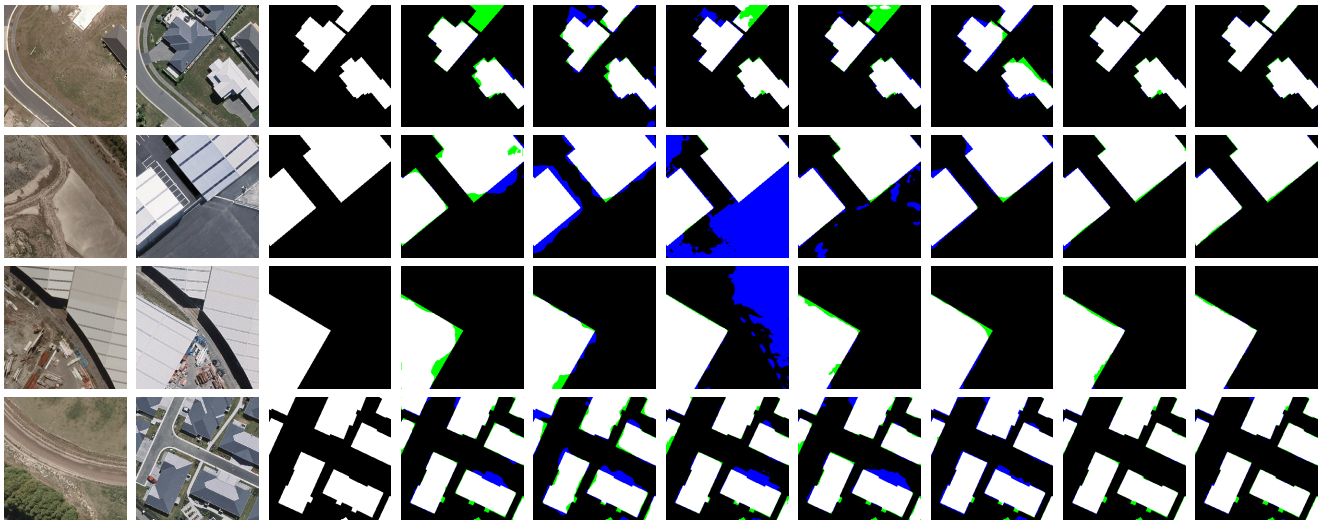


Fig. 6. Visualization results of WHU-CD. (a) and (b) Original bitemporal images. (c) Ground truth. (d)–(i) Prediction result of DSIFN, STANet, SNUNet, BIT, DSAMNet, and P2V-CD. (j) Prediction result of FMCD. In the resulting images, green pixels represent false negatives, blue pixels represent false positives, white pixels represent true positives, and black pixels represent true negatives.

Chongqing, Wuhan, and Xi'an. The authors of the dataset crop the data into image patches of size 512 × 512. After data augmentation, it is divided into 3600 pairs of images as training sets, 340 pairs of images as validation sets, and 48 pairs of images as test sets. In our experiment, according to the original data division, the images are cut into 256 × 256 sizes, and the number of the final training set, validation set, and test set are 14 400/1360/192, respectively.

Four common change detection evaluation indexes are selected to measure the performance of change detection methods, namely, precision (Pre), recall (Rec), overall accuracy (OA), and $F1$-Score [25]. The calculation process of metrics is as follows:

$$\begin{aligned} \text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Rec} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}} \\ F1 &= \frac{2}{\text{Rec}^{-1}+\text{Pre}^{-1}} \end{aligned} \tag{23}$$

where TP, FP, TN, and FN represent the number of true positives, false positives, true negatives, and false negatives, respectively.

B. Implementation Details

The FMCD is implemented on the Pytorch framework and trained on a single NVIDIA GeForce RTX 3090 GPU. AdamW was chosen to optimize the model, and the weight decay is set to 0.01. The initial learning rate is set to 0.001, and with epoch increases, the learning rate gradually decays through the cosine annealing function. Considering the limitation of memory, the BatchSize in the training process is set to 32. In the process of model training, data is enhanced by flipping, rotating, and adding random Gaussian noise, randomly. And the data enhancements for comparison methods are the same as the default settings mentioned in their papers.

To verify the superiority of the FMCD, six state-of-the-art models are compared: DSIFN [10], DSAMNet [14], STANet [16], SNUNet [15], BIT [25], and P2V-CD [20]. These comparison models achieve change detection in different ways and have made progress. Among them, DSIFN [10] extracts representative features through deep supervision and DSAMNet [14] combines metric learning and deep supervision on the basis of Siamese networks. At the same time, STANet [16] and SNUNet [15] achieve good results by introducing attention mechanisms, so they are chosen for comparison. Where STANet obtains long-range information through the self-attention mechanism, and the latter introduces channel attention to obtain the features of each level in the decoder. BIT [25] is a method based on the transformer, which uses semantic tokens to reduce the calculation amount of the self-attention mechanism. P2V-CD [20] extracts the temporal change features of bitemporal images in a video understanding way, which achieves good results.

C. Experimental Results on Three Datasets

In this section, the qualitative and qualitative evaluation of the experiment are analyzed as follows.

1) Qualitative Evaluation: The visualization results of the comparison methods on three datasets are shown in Figs. 6–8. Several colors are used to represent true positives (white), true negatives (black), false positives (blue), and false negatives (green). From Figs. 6–8, it can be seen that due to the different shooting times, the light, climate, and other factors cannot be consistent when shooting, resulting in different styles and tones of the two temporal images. This difference is particularly evident on the WHU-CD. Macroscopically, it can be seen that the DSIFN, SNUNet, and BIT are generally effective, but when the detection area has complex ground object information, such as the first line of Fig. 6, the second

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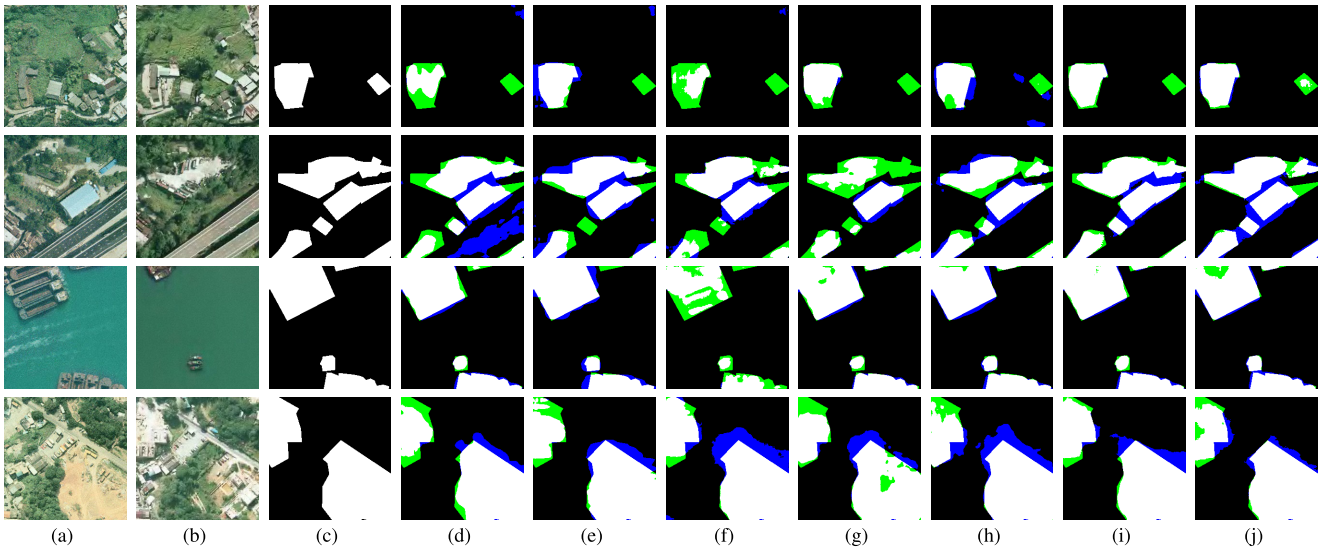


Fig. 7. Visualization results of SYSU-CD. (a) and (b) Original bitemporal images. (c) Ground truth. (d)–(i) Prediction result of DSIFN, STANet, SNUNet, BIT, DSAMNet, and P2V-CD. (j) Prediction result of FMCD. In the resulting images, green pixels represent false negatives, blue pixels represent false positives, white pixels represent true positives, and black pixels represent true negatives.

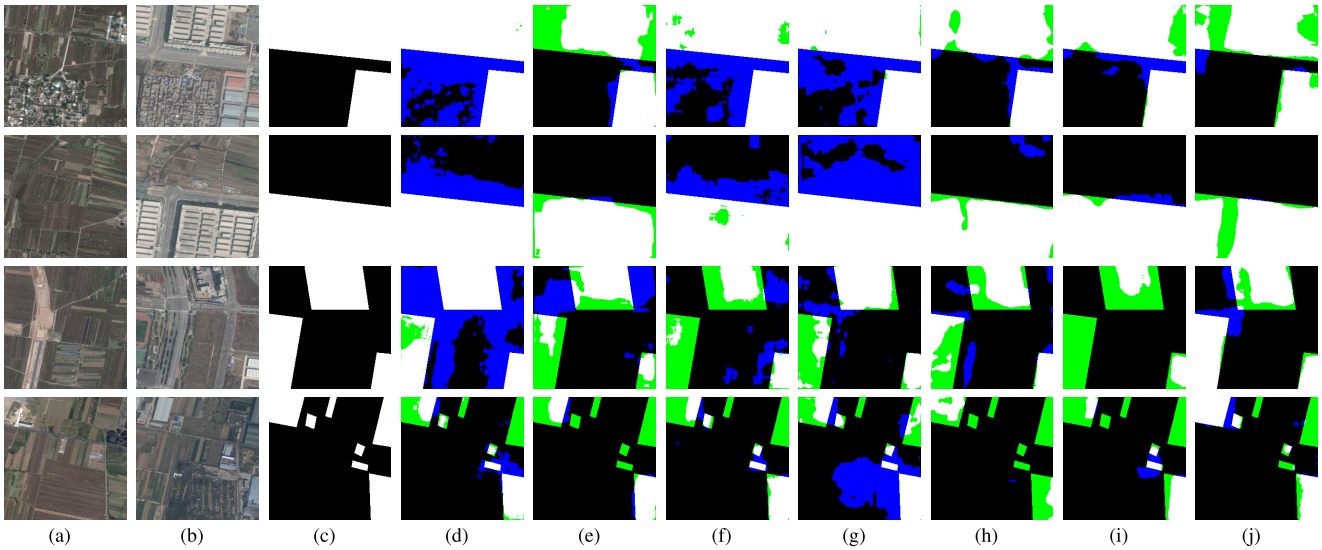


Fig. 8. Visualization results of DSIFN-CD. (a) and (b) Original bitemporal images. (c) Ground truth. (d)–(i) Prediction result of DSIFN, STANet, SNUNet, BIT, DSAMNet, and P2V-CD. (j) Prediction result of FMCD. In the resulting images, green pixels represent false negatives, blue pixels represent false positives, white pixels represent true positives, and black pixels represent true negatives.

line and the fourth line of Fig. 7, and the first line of Fig. 8, the error rate is significantly improved. In addition, STANet is less effective in the boundary part of the changing area. However, FMCD interacts with multilevel features in the multilevel feature interaction module, which combines the texture information in low-level features and the semantic information in high-level features. Therefore, the results of FMCD in complex scenes are better than other methods. It can be seen that the false alarm of FMCD is low, but the miss rate is high, that is, there are still many false negatives. Specific to the picture, it can be seen that most of the false negatives exist in areas where the definition of change is not obvious, such as the upper left in the third row of Fig. 7(j) and the lower left in the second row of Fig. 8(j). This is probably because in the feature distribution adaptive module, the model shorten the distance between the regions with the unclear definition of changes in the two temporal images, which leads to the difficulty of detecting these regions. However, the precision of detection is greatly improved by aligning the distribution of the reconstructed images to map the features of the two temporal images to the same space. From the lower right corner of the first line of Fig. 6, it can be seen that due to the inconsistent shooting conditions, the shadow appears around the house, which has a certain impact on the detection. Other methods have more or less deviation in the detection of house boundaries due to the presence of shadows. However, FMCD aligns the feature relationship between the two temporals, reducing the interference of shadow and other factors, thereby further improving the accuracy.

2) Quantitative Evaluation: Test results of different methods on these three datasets are shown in Tables I–III. The best and second best results of the metrics are highlighted in red and blue, respectively. Quantitative results show that FMCD has a good performance with a limited number of parameters,

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TABLE I
COMPARISON RESULTS ON WHU-CD DATASET

Method	Params(*M*)	OA	F1-Score	Pre	Rec
DSIFN	50.71	0.9882	0.8664	0.9222	0.8169
STANet	16.97	0.9871	0.8587	0.8369	0.8816
SNUNet	12.03	0.9715	0.7414	0.6198	0.9225
BIT	3.04	0.9885	0.8731	0.8536	0.8935
DSAMNet	16.95	0.9848	0.8840	0.7756	0.9256
P2V-CD	5.42	0.9944	0.9356	0.9518	0.9200
FMCD	3.20	0.9952	0.9448	0.9602	0.9299

TABLE II
COMPARISON RESULTS ON SYSU-CD DATASET

Method	Params(*M*)	OA	F1-Score	Pre	Rec
DSIFN	50.71	0.8950	0.7645	0.8115	0.7226
STANet	16.97	0.8818	0.7643	0.7212	0.8128
SNUNet	12.03	0.8947	0.7738	0.7843	0.7636
BIT	3.04	0.8992	0.7807	0.8019	0.7606
DSAMNet	16.95	0.8760	0.7559	0.7057	0.8139
P2V-CD	5.42	0.9036	0.7886	0.8166	0.7625
FMCD	3.20	0.9137	0.8114	0.8372	0.7872

TABLE III
COMPARISON RESULTS ON DSIFN-CD DATASET

Method	Params(*M*)	OA	F1-Score	Pre	Rec
DSIFN	50.71	0.8404	0.6249	0.5203	0.7821
STANet	16.97	0.8803	0.5298	0.7970	0.3968
SNUNet	12.03	0.7791	0.6249	0.4102	0.6855
BIT	3.04	0.8783	0.6445	0.6399	0.6492
DSAMNet	16.95	0.8667	0.5805	0.6241	0.5426
P2V-CD	5.42	0.8913	0.6019	0.7974	0.4834
FMCD	3.20	0.8868	0.6700	0.6641	0.6760

which is superior to other methods. On three datasets, for example, FMCD improved the $F1$-score by 0.92%, 2.28%, and 2.55%, respectively. Among them, the change of buildings is the main change type of WHU-CD, and the performance of FMCD on WHU-CD is the most obvious. The reason may be that FMCD uses multilevel feature interaction to aggregate the context information of features. It can be seen that the results of each algorithm on the DSIFN-CD are generally low, probably because DSIFN-CD are taken from different cities and the type of change varies greatly between cities. But unlike several other algorithms, such as SNUNet and STANet, FMCD makes a good trade-off between Pre and Rec, resulting in the highest $F1$-score. Through the transformer, BIT extracts the long-range semantic features of the images, which contain rich global spatial information and achieves good results. P2V-CD processes the bitemporal image from the perspective of video, which highlights the changing area. However, these methods do not consider the domain shift of bitemporal images caused by other factors such as illumination. In this respect, FMCD has achieved good results by setting up domain adaptive auxiliary tasks.

D. Ablation Studies

In this part, the influence and effect of some important modules and the multitask strategy of the FMCD are discussed.

TABLE IV
ABLATION EXPERIMENTS OF MULTILEVEL FEATURE INTERACTION MODULE (UNDER $F1$-SCORE)

	WHU-CD	SYSU-CD	DSIFN-CD
w/o	0.9322	0.8033	0.6476
FMCD	0.9448	0.8114	0.6700

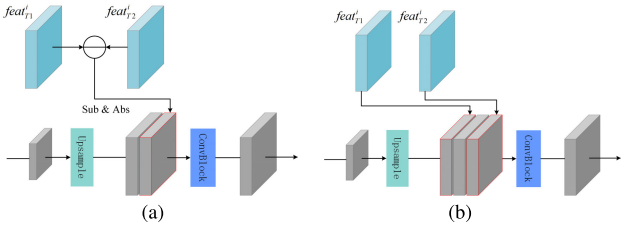


Fig. 9. Illustration of the way of skip connection. (a) Difference. (b) Concatenation.

First, the role of the multilevel feature interaction module and MAB in the change detection task is analyzed. In FMCD, the strategy of adaptively adjusting the loss weight of change detection tasks and domain adaptation tasks is adopted. Therefore, in this section, the comparative experiment of adaptively adjusting loss weights and fixed loss weights is implemented. The ablation experiments are carried out on three datasets. Due to the fact that $F1$-score can evaluate the comprehensive performance of the model, $F1$-score is selected as the index of the ablation experiment.

1) Effectiveness of the Multilevel Feature Interaction Module: To verify the effectiveness of the multilevel feature interaction module in FMCD, ablation experiments on this module are implemented. Since the multilevel feature interaction module is a necessary module to construct the whole framework, the outputs from Backbone are directly input into the change detection task and the domain adaptation task during ablation experiments.

In Table IV, w/o indicates that the multilevel feature interaction module is not used, and origin represents the original model of FMCD. Through the experimental results on three datasets, it can be obtained that the accuracy of the model is effectively improved after the module is adopted, which proves that the aggregation of low-level features and high-level features can obtain more representative features when processing images.

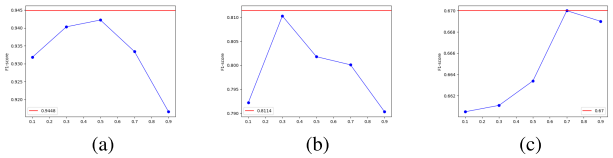
2) Effectiveness of MAB: In the change detection task, a novel block of skip connection is proposed, namely MAB. At present, most methods in change detection use concatenation [17], [49] or difference [2], [50] to obtain shallow features through skip connection. However, the way of difference will lose a lot of information, and the way of concatenation cannot make full use of the spatial-spectral joint features contained in the shallow features. Therefore, the way of skip connection has done the ablation experiment in this section.

In Table V, w/o means that no skip connections are used, Difference means that the difference maps of the shallow features are obtained and then the difference maps are concated into the upsampling process. Concatenation means that the shallow features obtained are directly concated into the

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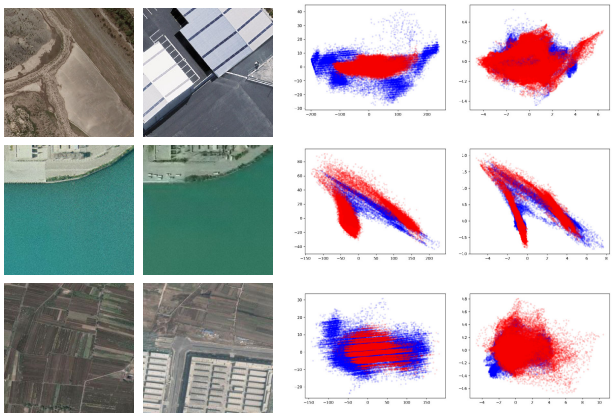
TABLE V
ABLATION EXPERIMENTS OF SKIP CONNECTION (UNDER $F1$-SCORE)

Skip Connection	WHU-CD	SYSU-CD	DSIFN-CD
w/o	0.9342	0.8001	0.6588
Difference	0.9419	0.8097	0.6427
Concatenation	0.9376	0.8055	0.6603
MAB	0.9448	0.8114	0.6700



(a) (b) (c)

Fig. 10. Experimental results under different fixed loss weights. Where the red line represents the experimental results under the strategy of adaptive adjusting loss weight and the blue line represents the experimental results under different fixed loss weights. (a) WHU-CD. (b) SYSU-CD. (c) DSIFN-CD.



(a) (b) (c) (d)

Fig. 11. Data distribution map of bitemporal images before and after domain adaptation. (a) and (b) Two temporal images. (c) Distribution of the two temporal images before domain adaptation. (d) Distribution of the two temporal images after domain adaptation.

upsampling process. As shown in Fig. 9, the difference and the concatenation are mainly due to the different fusion methods of the low-level features from the Backbone, where the ConvBlock is two convolutional layers with kernel sizes of 3 and 1, respectively. The difference between the two ConvBlocks is mainly in the input channels. Through experiments, skip connections can introduce features with rich low-level information, which is important for dense prediction tasks [51]. Both differences and concatenation have advantages for different datasets, but neither can make full use of the information contained in the shallow features. The way of adopting MAB in skip connections has the highest $F1$-score, indicating that MAB can effectively interact and aggregate features between two temporal images.

3) Effectiveness of Adaptively Adjusting Loss Weight: In FMCD, the strategy of adaptively adjusting the loss weight through homoscedasticity uncertainty is adopted. The experimental results under this strategy are compared with the experimental results under the loss of different fixed weights. A general way to measure the importance of the loss of two tasks, according to formula (2), is adopted. The values of α are 0.1, 0.3, 0.5, 0.7, and 0.9, respectively, and experiments are carried out on three datasets.

As shown in Fig. 10, because different datasets have different characteristics, the optimal value of α is also different. The highest $F1$-Score can be obtained on WHU-CD, SYSU-CD, and DSIFN-CD when α is 0.5, 0.3, and 0.7, respectively. In addition, the blue line does not exceed the red line, which shows that the strategy of adaptively adjusting loss weight is superior to the fixed loss weight to some extent. And it also shows that the domain adaptive task has a certain auxiliary effect on the change detection task.

V. DISCUSSION

FMCD sets the domain adaptation between two temporal images as an auxiliary task for change detection. Therefore, the role of the domain adaptation task for the overall model is discussed in this section. In addition, the experiment of exploring the function of the result of the domain adaptation task is further implemented.

A. Effectiveness of Domain Adaptation

1) Reconstruction Results in Domain Adaptation Tasks: In the domain adaptation task of FMCD, the purpose of reconstruction is to map the representative features of the two-phase images obtained by the encoder to the same feature space. Therefore, in the process of reconstruction, the reconstruction of the original image is not the main work. The main work of the domain adaptive module in FMCD is to narrow the feature space of the source domain and the target domain by reconstruction.

Since both the reconstructed image and the original image are color images with three channels, to facilitate the observation of their data distribution, PCA dimensionality reduction technology is applied. Each pixel of the three-channel image is converted into 2-D data by PCA and then drawn into a scatter plot. In Fig. 11(c) and (d), the red part represents the data distribution of the first temporal image, and the blue part represents the data distribution of the second temporal image. The image style of the first row and the third row in Fig. 11 is quite different, and there are many change areas, so the data distribution of them is discrepant. The difference between the second line of the images is mainly in the tone or style, so after the domain adaptation, the data distribution map of the two images is close to coinciding. It can be seen from the figure that the data distribution areas of the two images tend to be similar after domain adaptation, which indicates that the data distribution of the bitemporal image is more uniform.

2) Effectiveness of Feature Relationship Adaptation Module (FRAM) and Feature Distribution Adaptation Module (FDAM): In the domain adaptation task, two modules for feature alignment are conducted ablation experiment, which are the FDAM and the FRAM.

As shown in Table VI, the feature relationship adaptive module and the feature distribution adaptive module both improve the performance of the model more or less. The

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TABLE VI
ABLATION EXPERIMENTS OF DOMAIN ADAPTATION (UNDER $F1$-SCORE)

FDAM	FRAM	WHU-CD	SYSU-CD	DSIFN-CD
		0.9315	0.8007	0.6573
	✓	0.9402	0.8069	0.6675
✓		0.9372	0.8039	0.6610
✓	✓	0.9448	0.8114	0.6700

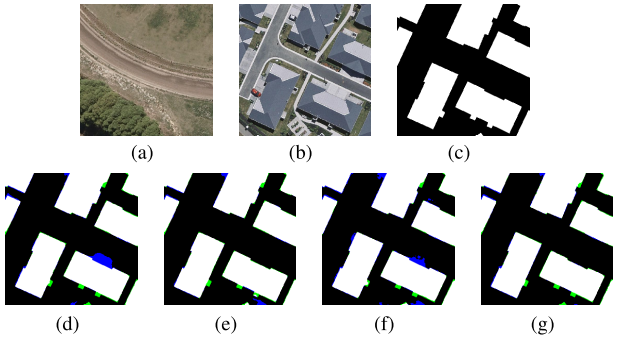


Fig. 12. Visualization results of ablation experiments of each domain adaptive module in FMCD on WHU-CD. (a)–(c) T1 image, T2 image, and ground-truth map. (d) Visual results without domain adaptation task. (e) FRAM only. (f) FDAM only. (g) Complete FMCD.

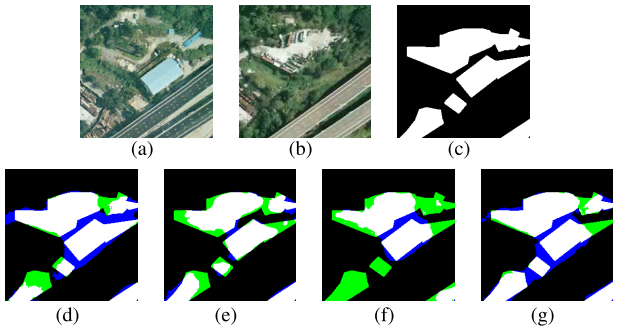


Fig. 13. Visualization results of ablation experiments of each domain adaptive module in FMCD on SYSU-CD. (a)–(c) T1 image, T2 image, and ground-truth map. (d) Visual results without domain adaptation task. (e) FRAM only. (f) FDAM only. (g) Complete FMCD.

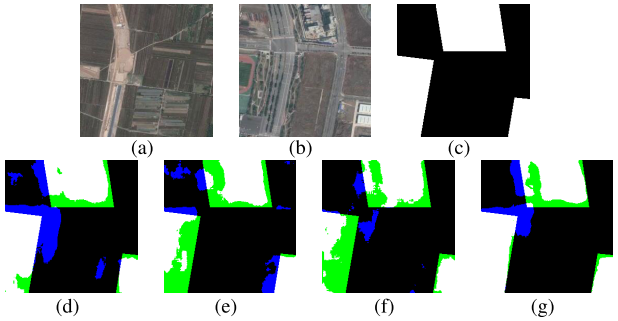


Fig. 14. Visualization results of ablation experiments of each domain adaptive module in FMCD on DSIFN-CD. (a)–(c) T1 image, T2 image, and ground-truth map. (d) Visual results without domain adaptation task. (e) FRAM only. (f) FDAM only. (g) Complete FMCD.

feature relationship adaptive module improves the performance of the model more obviously. The reason may be that the feature relationship adaptive module aligns the relationship between the features of the two temporal images at the global level, so some irrelevant changes are ignored, making the boundary between the changing region and the invariant region more obvious. Visualization results of ablation experiments on three datasets are shown in Figs. 12–14. When using the feature distribution adaptive module alone, the false positives are significantly reduced, while the false negatives are increased. The reason is probably that by aligning the data distribution of the reconstructed results, the module maps the two temporal feature maps in the same space, reducing the impact of excessive differences in feature distribution, but this will affect the detection of the changing area to a certain extent. When the two modules work simultaneously, FMCD can not only distinguish the boundary between the changed region and the unchanged region, reduce the interference of irrelevant changes, but also map the features of the two temporals to the same space, making it easier for the model to learn to detect changes.

TABLE VII
EXPERIMENT OF FUSING THE RESULTS OF THE TWO TASKS (UNDER $F1$-SCORE)

	WHU-CD	SYSU-CD	DSIFN-CD
Fusion	0.9320	0.8071	0.6559
FMCD	0.9448	0.8114	0.6700

B. Further Exploration of Domain Adaptation

The results of domain adaptation still contain a lot of useful information, so whether the results after domain adaptation can be reused by change detection is discussed in this section. The images of two temporals after domain adaptive reconstruction has little difference in data distribution, so other factors such as image style have less interference. Taking this into account, the experiment of fusing the results of the two tasks is set up. The results of the reconstruction in the domain adaptation task are fused with the feature maps generated in the change detection task and then sent together to the classifier to detect changes.

In this experiment, the concatenation operation was used to fuse the results of the two tasks. The experimental results are shown in Table VII. And Fusion means that the two tasks are fused and sent to the classifier to obtain the final change detection result, and FMCD means the FMCD without any change. From the results, the fusion of the two tasks will weaken the performance of the model, resulting in a decrease in the results. The reason may be that the reconstruction decoder of the domain adaptive part in FMCD has a simple structure, which aims to constrain the feature encoder by reconstruction and map the features of two temporal images generated by the feature encoder into the same feature space. So it is difficult to obtain high-quality reconstructed images and when the domain adaptation task is fused with the change detection task, the reconstruction results will interfere with the original features.

VI. CONCLUSION

In this article, FMCD is proposed. FMCD improves the detection ability of the model in complex scenes by extracting representative features through the interaction of different levels of features in one temporal and the interaction between

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the features of two temporals. A multilevel feature interaction module is used to model the context information of different levels of features in a temporal. Then, the interaction and aggregation of features between two temporal images are realized by MAB, which improves the sensitivity to the changing area. At the same time, adopting a multitask learning mechanism, FMCD sets up a domain adaptive auxiliary task to reduce the influence of domain shifts and irrelevant changes caused by different weather, light, and other factors in bitemporal images. Experiments on three datasets show that FMCD outperforms most state-of-the-art change detection methods.

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